Lesson 30: Arc Length of Plane Curve and Parametric Curve

1) Let $f(x) = (x^2+1)^{3/2}$

Find the arc length of the graph of $f(x)$ on the interval $[0,2]$.

a) $f'(x) =$ ___?___

b) $s = \text{arc length} = \int_a^b \sqrt{1+[f'(x)]^2} \cdot dx =$ ___?___

2) Let $f(x) = \ln(3x^3+1)$

Find the arc length of the graph of $f(x)$ on the interval $[0,2]$.

a) $f'(x) =$ ___?___

b) $s = \text{arc length} = \int_a^b \sqrt{1+[f'(x)]^2} \cdot dx =$ ___?___

3) Let $f(x) = \dfrac{1}{x+1}$

Find the arc length of the graph of $f(x)$ on the interval $[0,2]$.

a) $f'(x) =$ ___?___

b) $s = \text{arc length} = \int_a^b \sqrt{1+[f'(x)]^2} \cdot dx =$ ___?___

4) Let $g(y) = 3y^2 + 2$

Find the arc length of the graph of $g(y)$ on the interval $y \in [0,2]$.

a) $g'(y) =$ ___?___

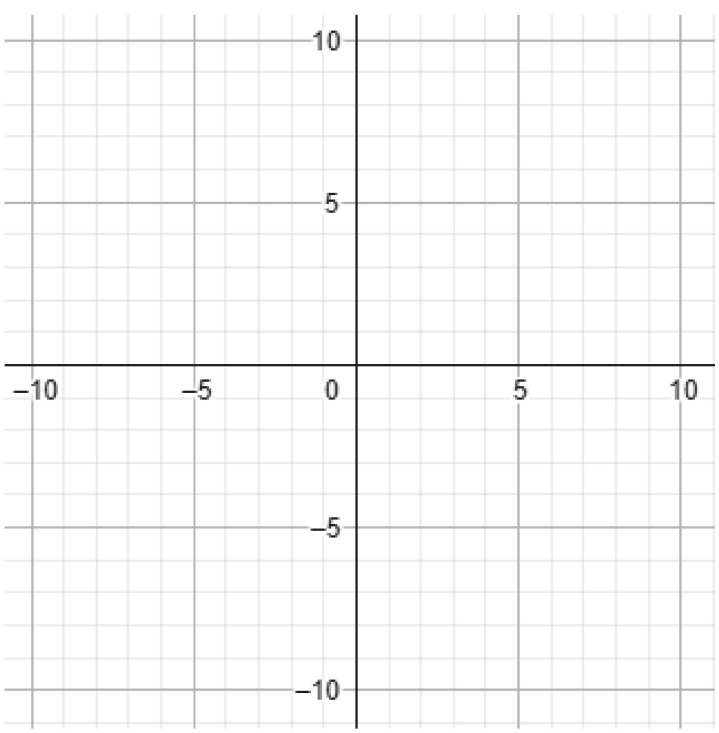
b) $s = \text{arc length} = \int_c^d \sqrt{1 + [g'(x)]^2} \cdot dy =$ ___?___

5) Parametric Equations: $x = 3t + 2,\ \ y = 4t \qquad 0 \le t \le 2$

a) Graph parametric equations $x = 3t + 2,\ \ y = 4t \qquad 0 \le t \le 4$

Hint: Use online 2D Calculator and input Curve(3t+2, 4t, t, 0, 4).

b) $s = \text{Arc Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$ ___?___

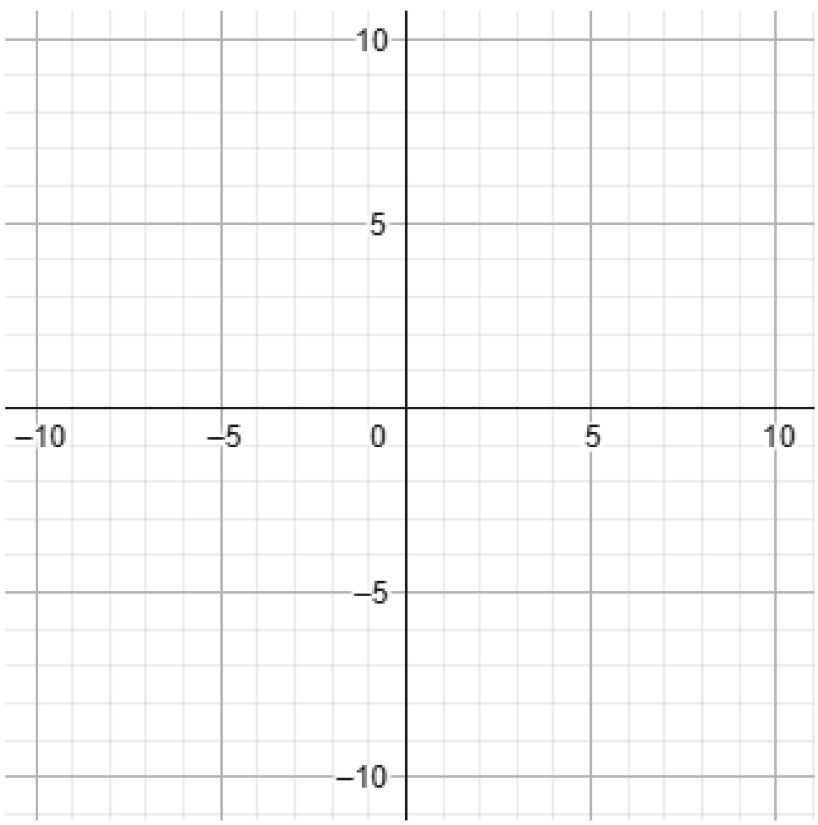


6) Parametric Equations: $x = t^2 + 2, \ \ y = t^3 \qquad 0 \le t \le 2$

a) Graph parametric equations $x = t^2 + 2, \ \ y = t^3 \qquad 0 \le t \le 2$

Hint: Use online 2D Calculator and input Curve(t^2+2, t^3, t, 0 , 2).

b) s = Arc Length = $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$?

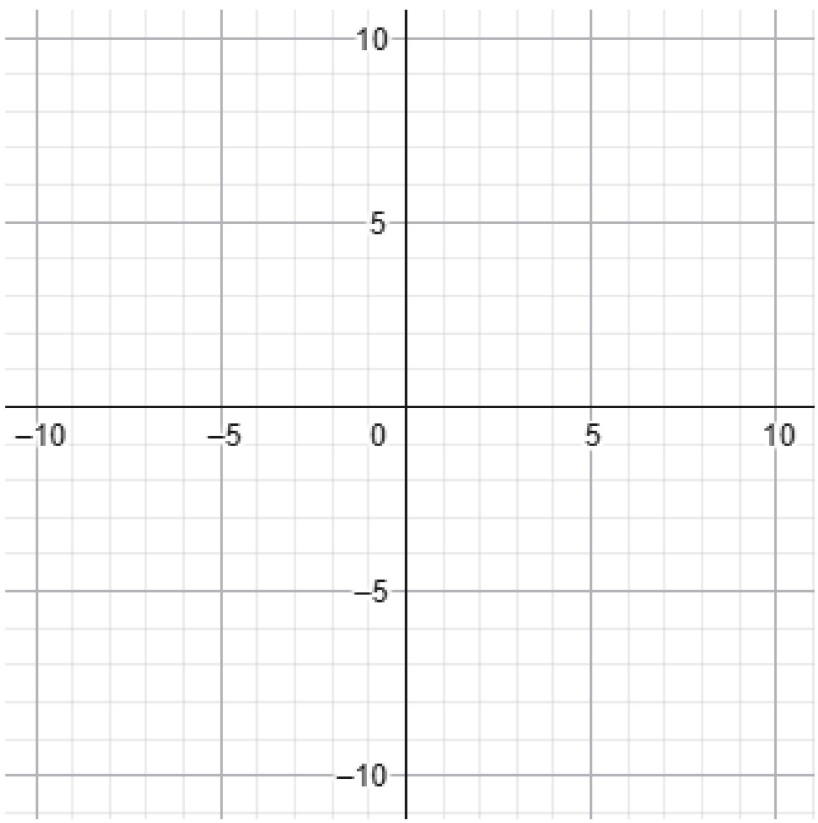


7) Parametric Equations: $x = 3\cos t + 2,\ \ y = 4\sin t \qquad 0 \le t \le \pi/3$

a) Graph parametric equations $x = 3\cos t + 2,\ \ y = 4\sin t \qquad 0 \le t \le \pi/3$

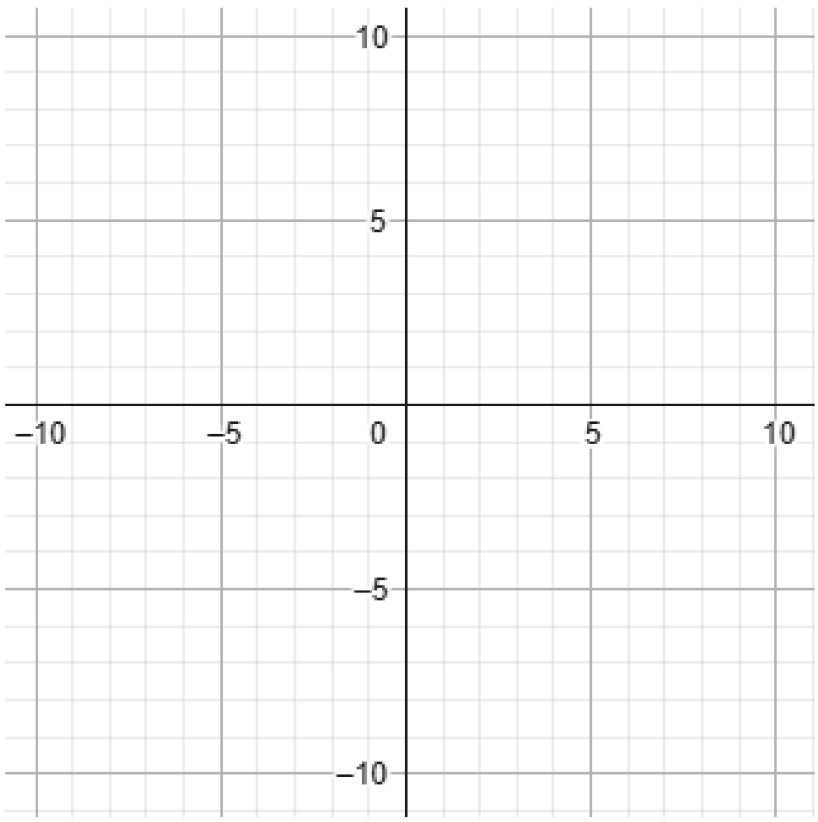
Hint: Use online 2D Calculator and input Curve(3cos t, 4sin t, t, 0 , pi/3).

b) $s = \text{Arc Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$?



8) Parametric Equations: $x = e^t$, $y = e^{2t}$ $\quad 0 \le t \le 1$

a) Graph parametric equations $x = e^t$, $y = e^{2t}$ $\quad 0 \le t \le 1$

Hint: Use online 2D Calculator and input Curve(e^t, e^2t, t, 0 , 1).

b) s = Arc Length = $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$?